

Interactive Computer Graphics

Chapter 10 - Shadows

Summer Term 2026

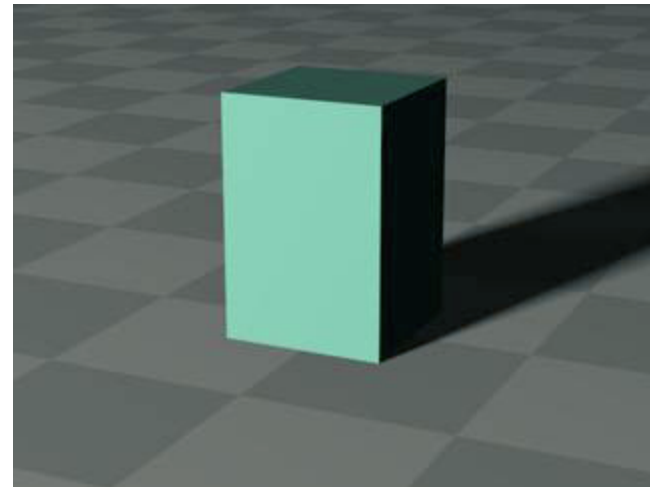
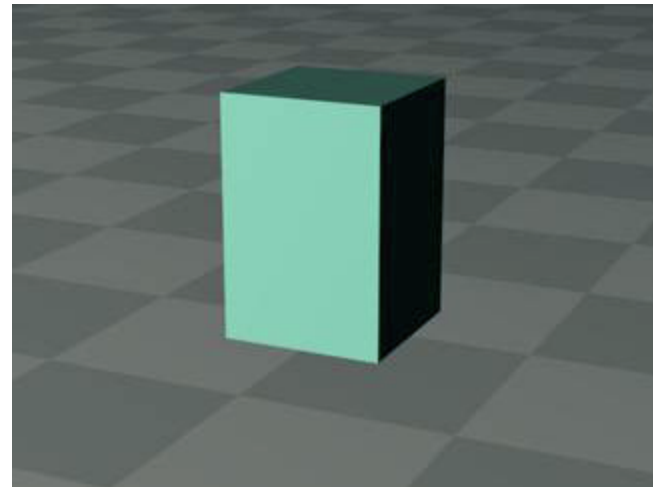
Timo Ropinski



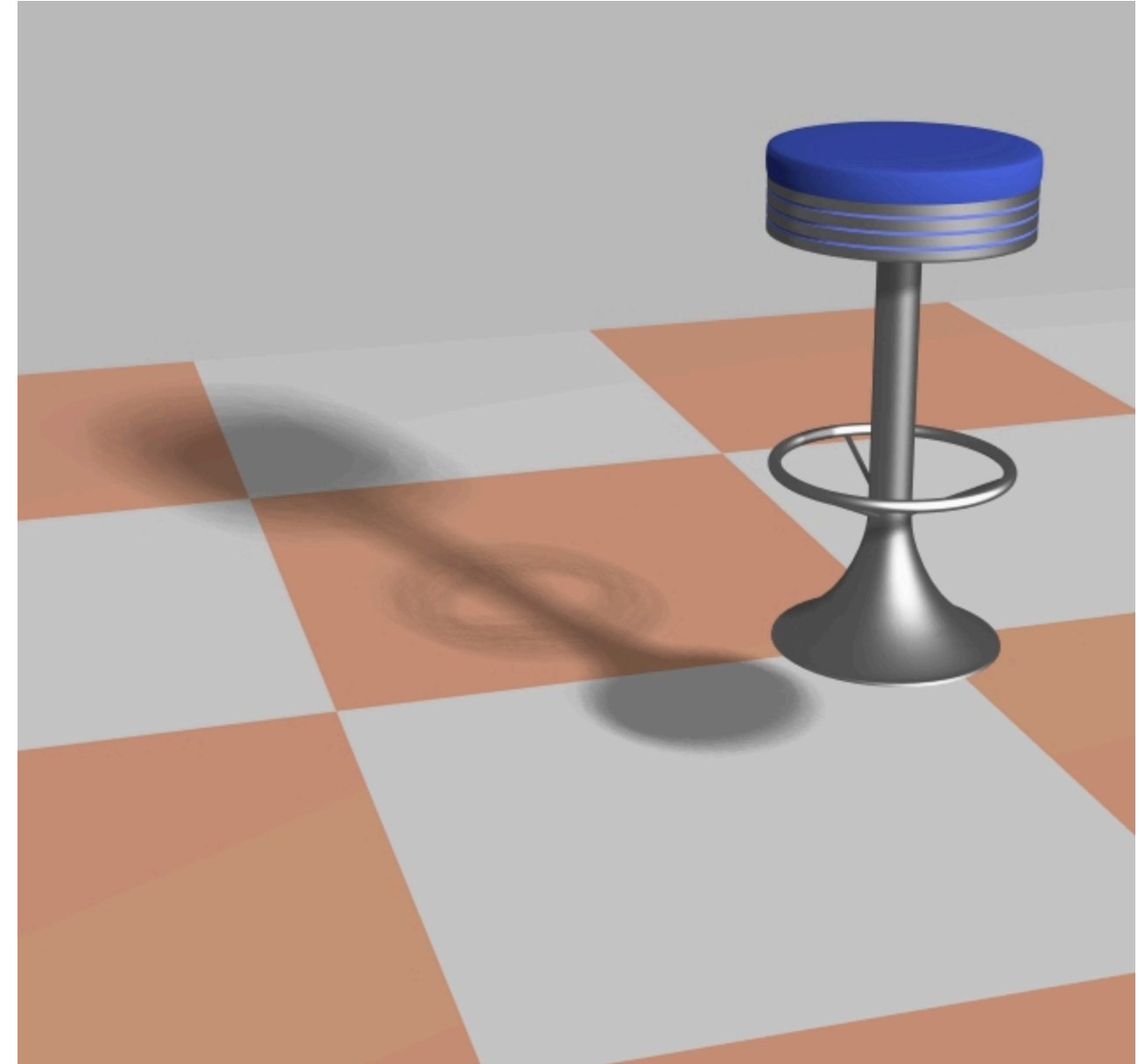
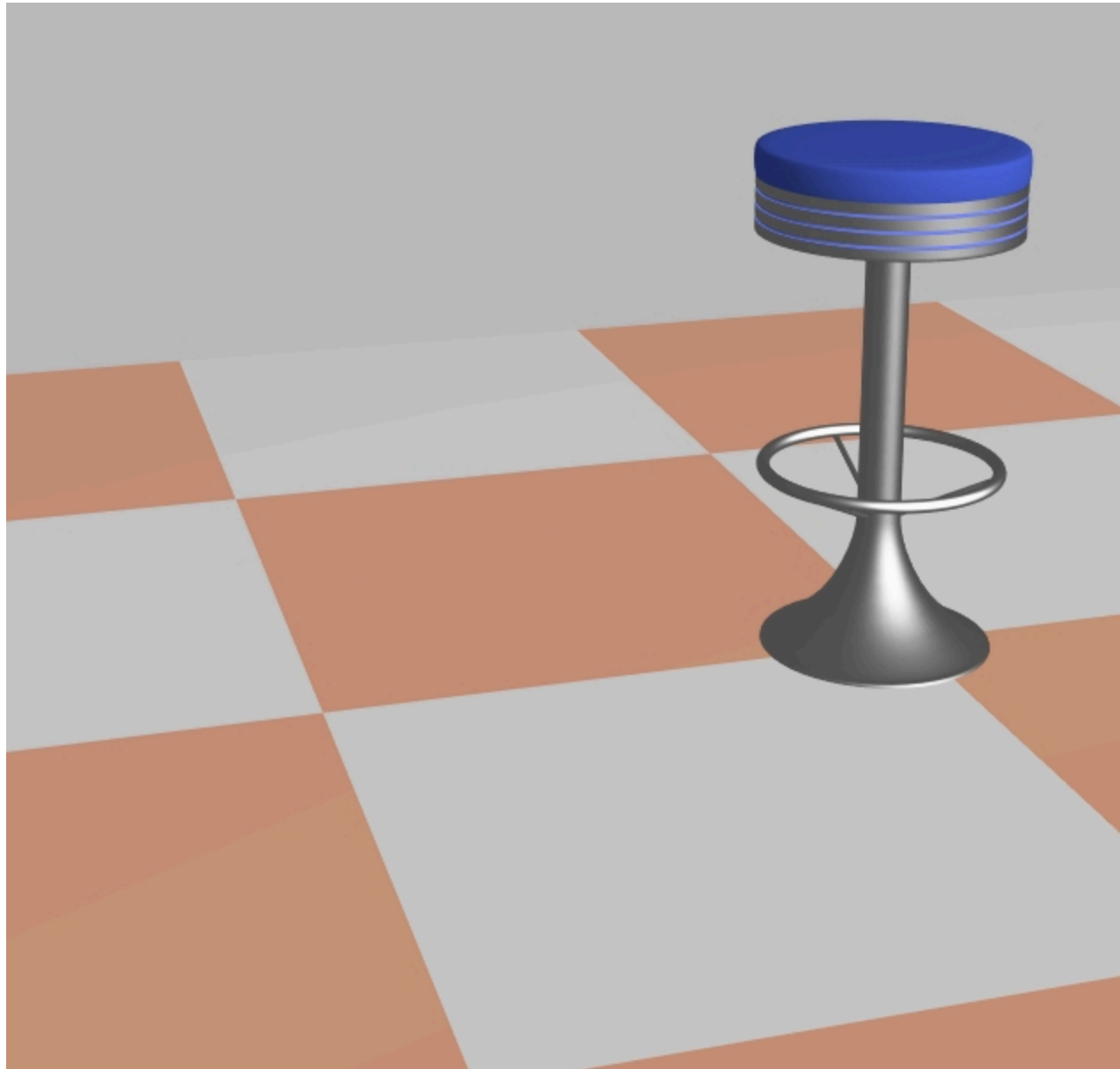
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Shadow Perception 1/3

- Visual perception of an environment is influenced by shadows
 - Object object relationships
 - Object space relationships
- Improved depth perception is made possible by shadows
- Shadows provide appealing visual effects



Shadow Perception 2/3



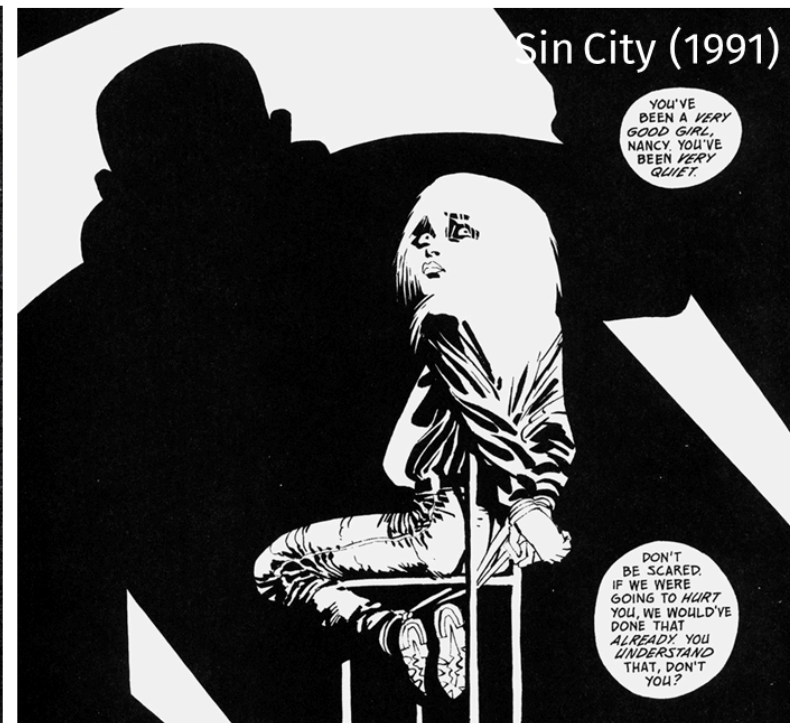
Shadow Perception 3/3

- Kersten D. Knill, D. C., Mamassian, P. and Bülthoff I. Illusory motion from shadows, Nature, 379 (31), 1996.

Shadow Atmosphere

Shadow Atmosphere

- Shadows create a certain atmosphere



- Shadows create a certain atmosphere
- The Lighthouse (2019)

Shadow Atmosphere

- Shadows create a certain atmosphere
- The Lighthouse (2019)



- Kill Bill: Volume 1 (2003)

Overview

10.1 Definitions

10.2 Ground Plane Shadows

10.3 Light Maps

10.4 Shadow Volumes

10.5 Shadow Maps

References

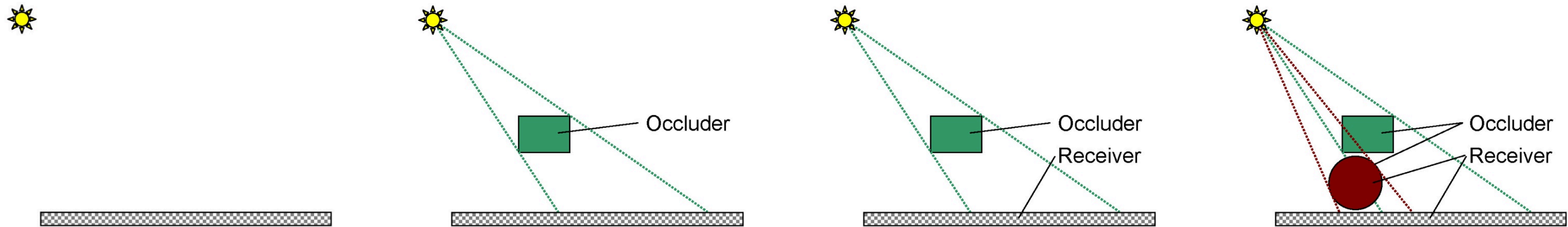
10.1 Definitions

About shadow-receiving and shadow-casting scene objects



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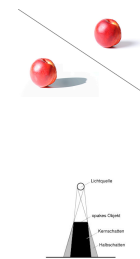
Classification of scene objects



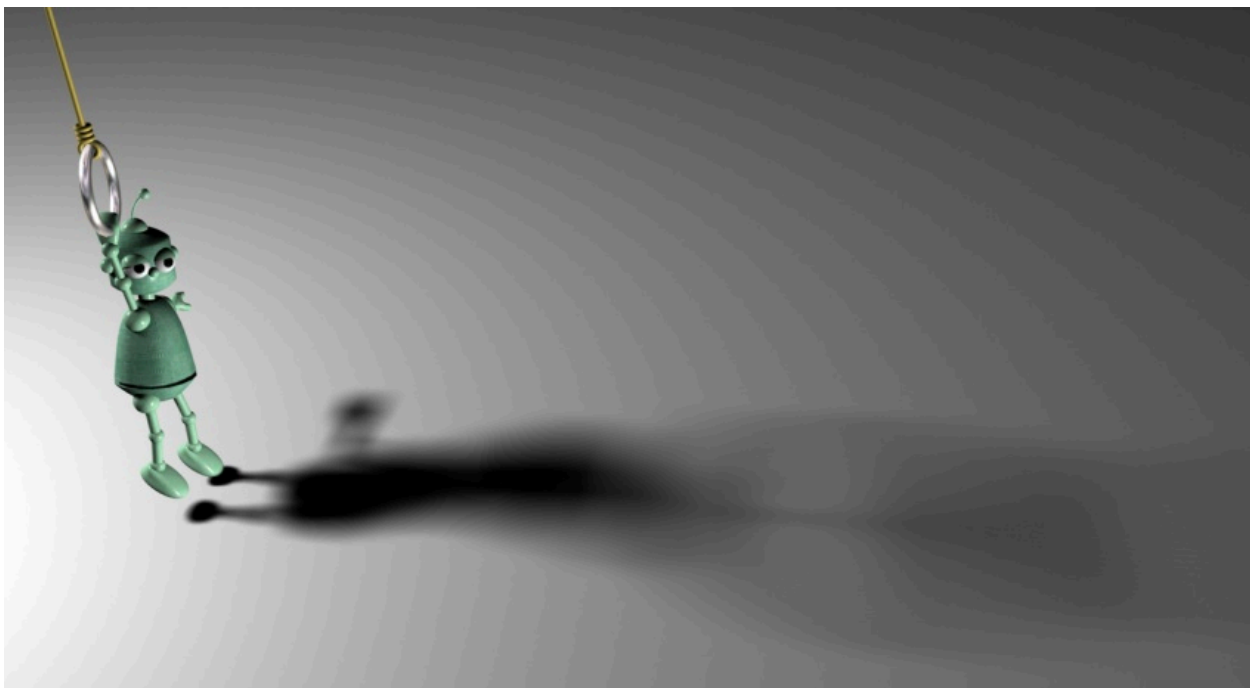
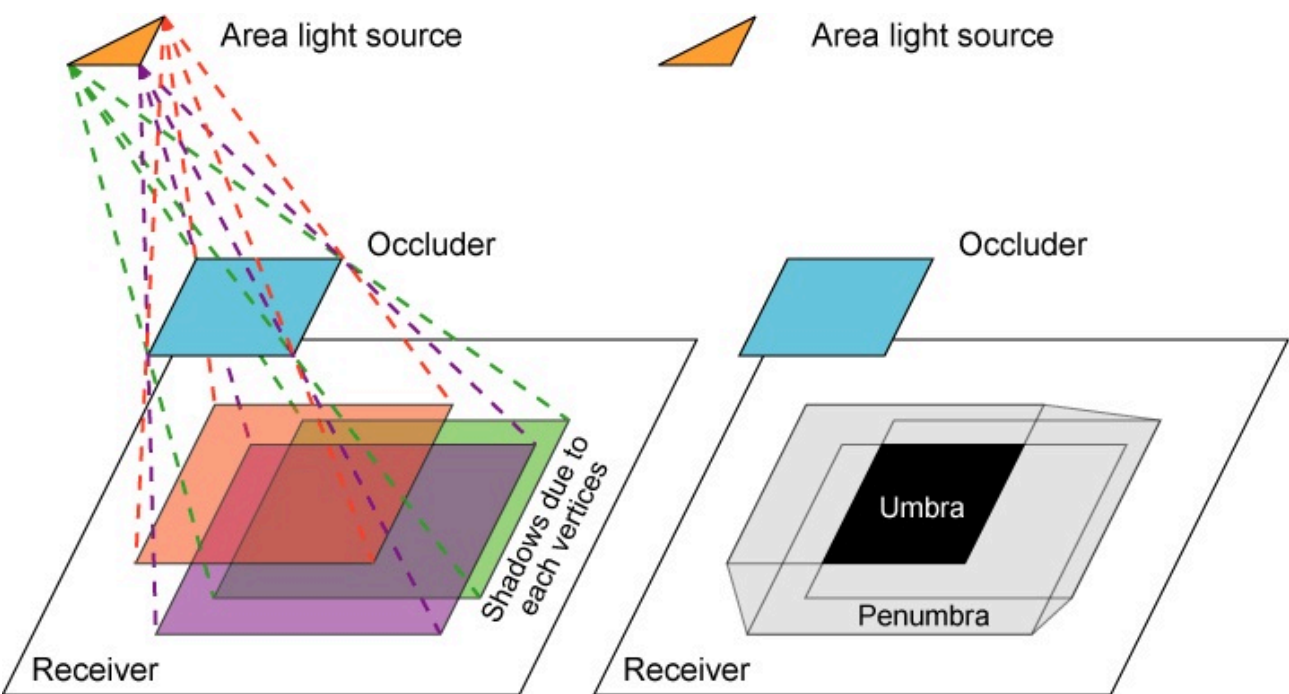
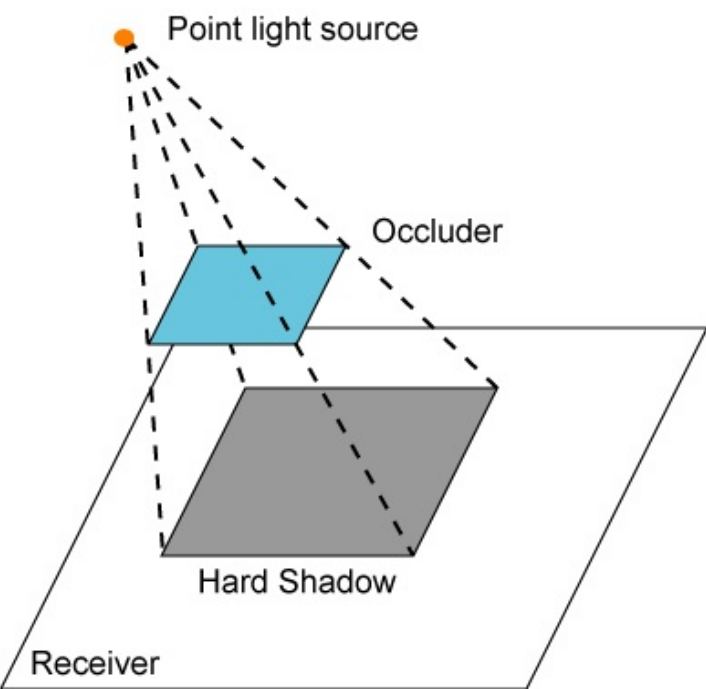
- Distinction between
 - Shadow-casting scene objects (Occluder)
 - Shadow-receiving scene objects (Receiver)
- Classify shadow-casting and shadow-receiving objects in a preprocess?
 - Self-shadowing results from same object casting and receiving shadows

Umbra and Penumbra

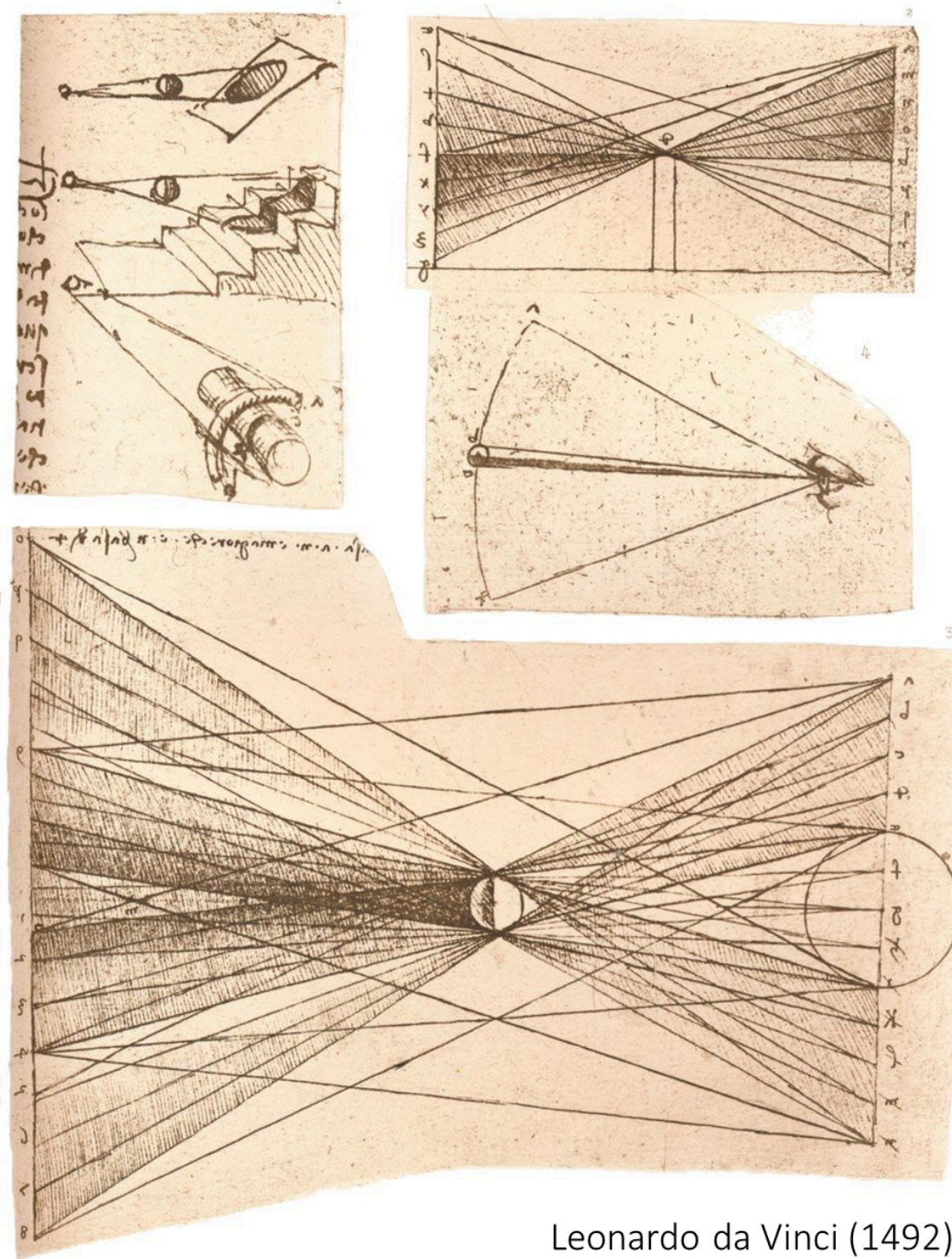
- Point light sources
 - Result in hard shadows
 - High intensity contrast between shadowed and non-shadowed region
- Area light sources
 - Result in soft shadows
 - Core shadow (umbra): area completely occluded wrt. light source
 - Penumbra (Penumbra): area only partially occluded wrt. light source
- Size of shadow region depends on
 - Area of light source
 - Distance of light source to scene object



Hard and Soft Shadows



Description of Shadows



Leonardo da Vinci (1492)

- Shadows can be described differently
- Shadows can be modeled as
 - Separate objects
 - Surfaces not seen when looking into scene from light source
 - Volumetric regions with reduced lighting

Observations 1/2

- Possible shadow generation approaches
 - Shadows as surfaces: exploit hidden-surface algorithms
 - Shadows as projection: create shadows from A to B by projecting A onto B from light source
- Static vs. dynamic scenes
 - Shadows are static in scenes with static lighting and scene objects (shadow computation possible as preprocess)
 - Shadows are dynamic in scenes with dynamic lighting and/or dynamic scene objects (shadows must be computed per frame)

Observations 2/2

- Shadow shape can be obtained by projection
 - Trivial for projection on flat surfaces
 - Algorithmically very complex for projection in general case
- Shadow intensity
 - Shadows attenuate the incoming light used in illumination model
- Correct shadows can only be computed with global illumination models
- Empirical approaches for shadow computation are widely used

Shadow Calculation Methods

- Ground plane shadows
 - Static shadow polygons
 - Dynamic shadow polygons
- Projected shadows
- Light maps
- Shadow volumes
- Shadow maps
- Interactive Computer GraphicsChapter 10 – Shadows
- 27/06/2026

10.2 Ground Plane Shadows

Static and dynamic shadow polygons

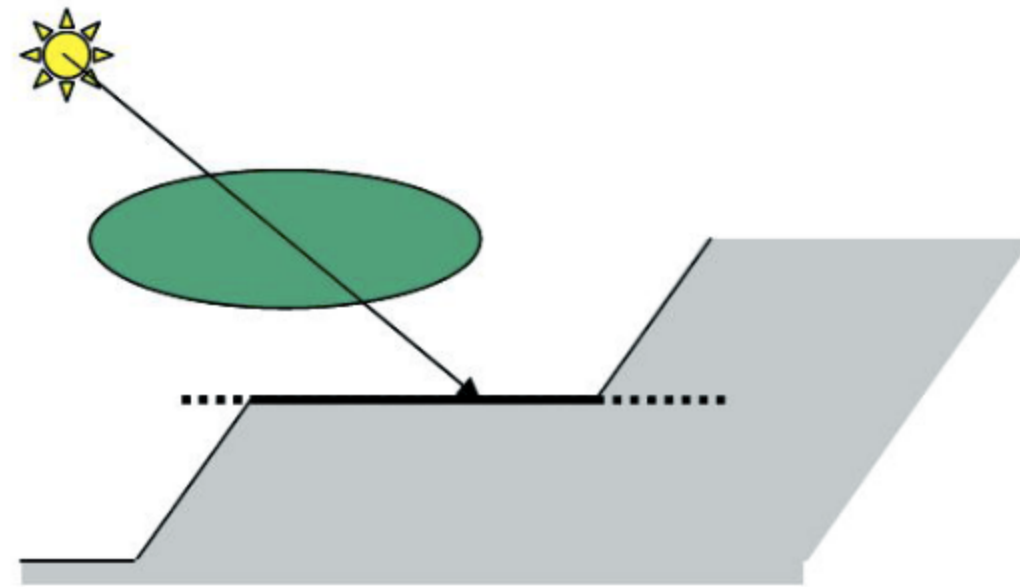


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Static Shadow Polygons

- Render dark polygons to represent shadow regions
 - Cast ray from light source through center of shadow-casting objects
 - Position dark polygon where ray intersects shadow-receiving object
- Disadvantages
 - Static shapes cannot represent geometry of dynamic scene objects
 - Single intersection of ray and shadow-receiver can't capture soft shadows
 - Incorrect parts of the shadow may be drawn

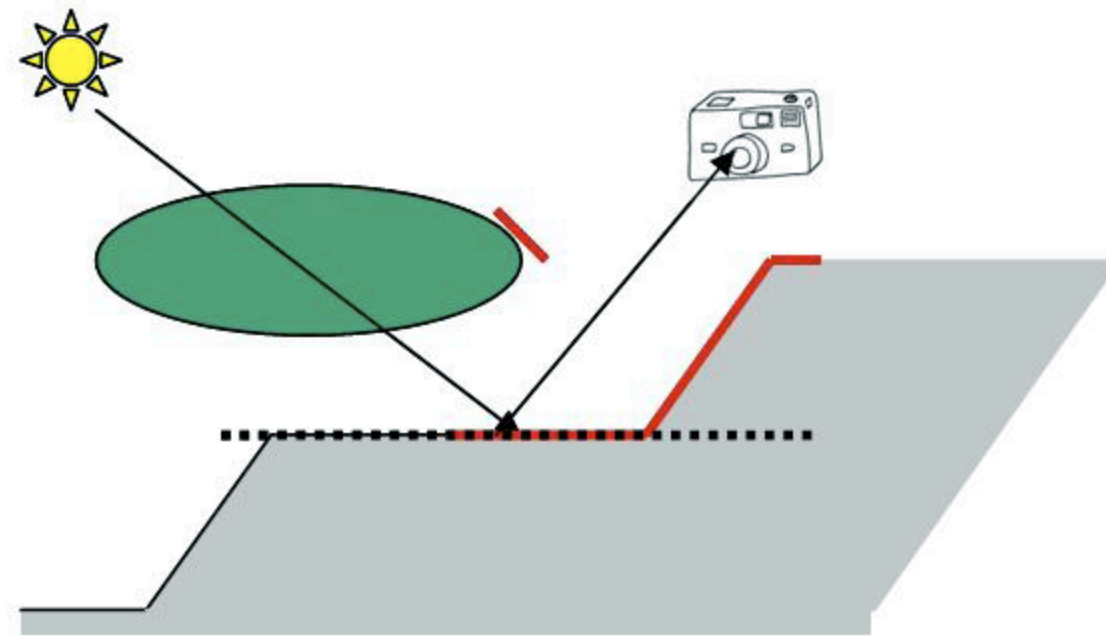
Render Static Shadow Polygons 1/2



- Depth value identical with that of shadow-receiving polygon (care must be taken wrt. z-buffer fighting)
- Depth value smaller than that of shadow-receiving polygon (shadow can be too large)

Render Static Shadow Polygons 2/2

- Disable depth test
 - Cast ray to determine shadow visibility
 - Show entire shadow polygon
- Shadow too large and possibly in wrong places
- Test corners for shadow visibility
 - Sufficient for planar polygons



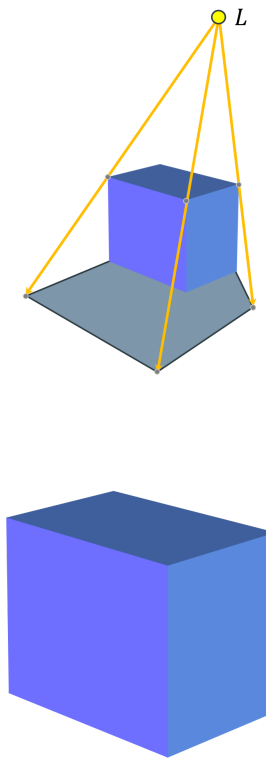
Static Shadow Polygons

- Good for car racing games (static scene geometry)
- With fast movements and proximity of casters and receivers, wrong shadows are often not detected as such



Dynamic Shadow Polygons

- Project shadow-casting scene objects onto shadow-receiving object by modifying shape
- Accurate procedure for semi-transparent shadows, but only if shadows do not overlap
- Restrictions
 - Only planar polygons on shadow-receiving scene objects possible
 - No self-shadowing possible



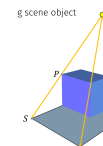
Shadow Projection

- Shadow point S (= vertex of the shadow polygon) lies on a line between light source L and vertex P of shadow-casting scene object

$$S = P - \alpha \cdot L$$

- S is located on base area with $z_S = 0$ with $\alpha = \frac{z_P}{z_L}$
 - $x_s = x_P - \frac{z_P}{z_L}x_L$ and $y_s = y_P - \frac{z_P}{z_L}y_L$
- In matrix form

$$[x_S y_S 0 1] = [1 0 - \frac{x_L}{z_L} 0 0 1 - \frac{y_L}{z_L} 0 0 0 0 0 0 0 1][x_P y_P z_P 1]$$



Rendering Dynamic Shadow Polygons

- Matrix M transforms object into its shadow polygon
- Render scene object with shadows
 - Render scene object
 - Multiply current model-view -matrix with M
 - Render scene object in shadow color with blending
- Possible extensions
 - Lift shadow polygons slightly to avoid z-fighting (integrate bias in M)

10.3 Light Maps

Structured lighting with textures

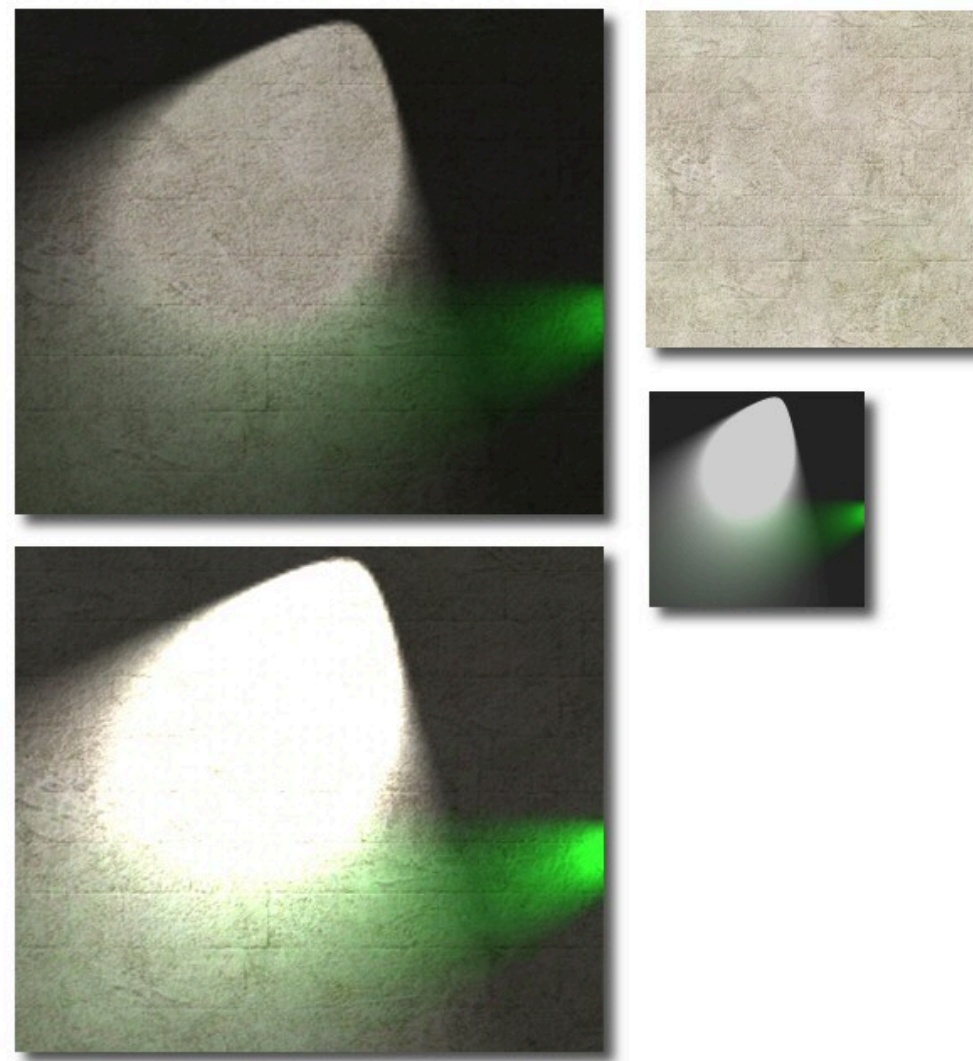


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Light Maps

- Static shadows can be precomputed and stored in light maps
- When precomputing light maps, shadows are calculated offline(e.g. with ray tracing)
- Soft shadows in lightmaps
 - Interpolation of the lightmap creates soft shadow approximations
 - Light source sampling results in better results,sending multiple rays to different points of the light source(additional filtering improves image quality)

Soft Shadows with Light Maps 1/2



Soft Shadows with Light Maps 2/2



10.4 Shadow Volumes

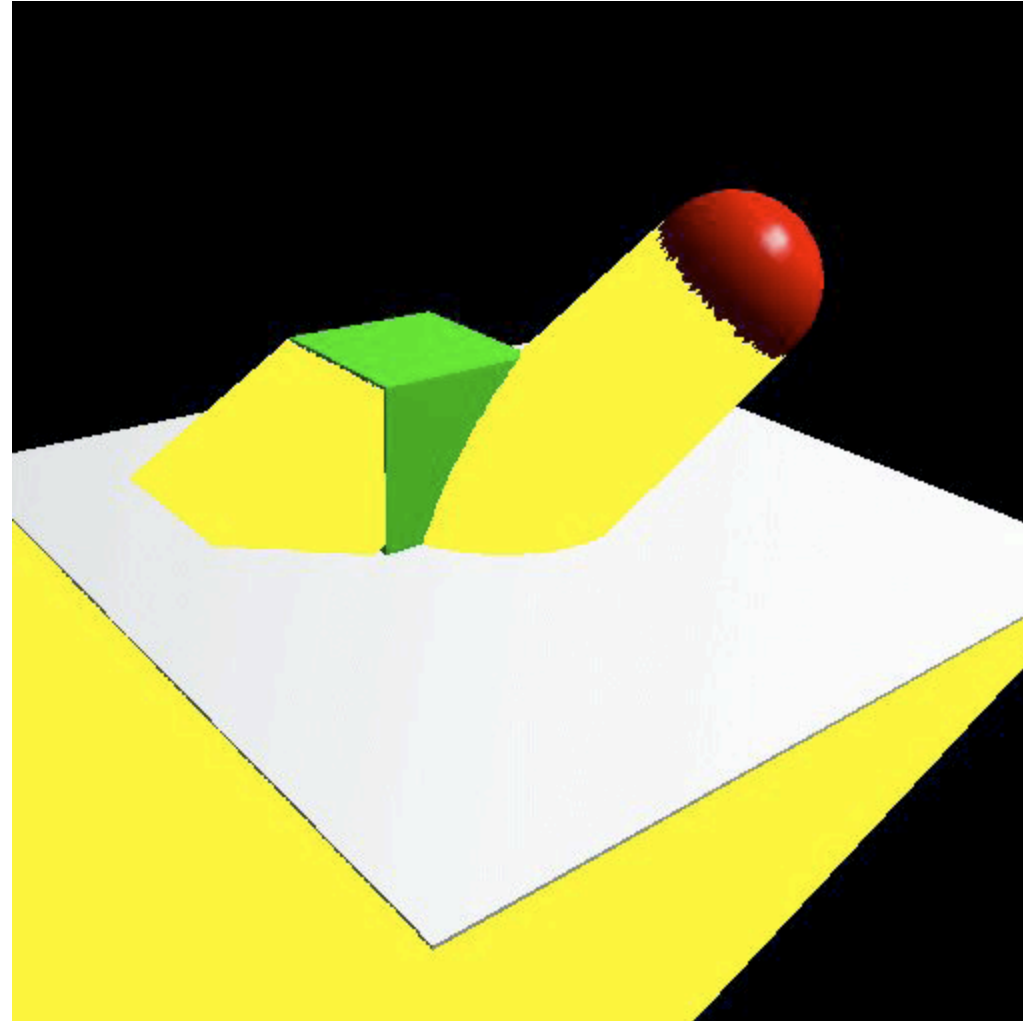
Representation of shadows by geometry enclosing them



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Shadow Volumes

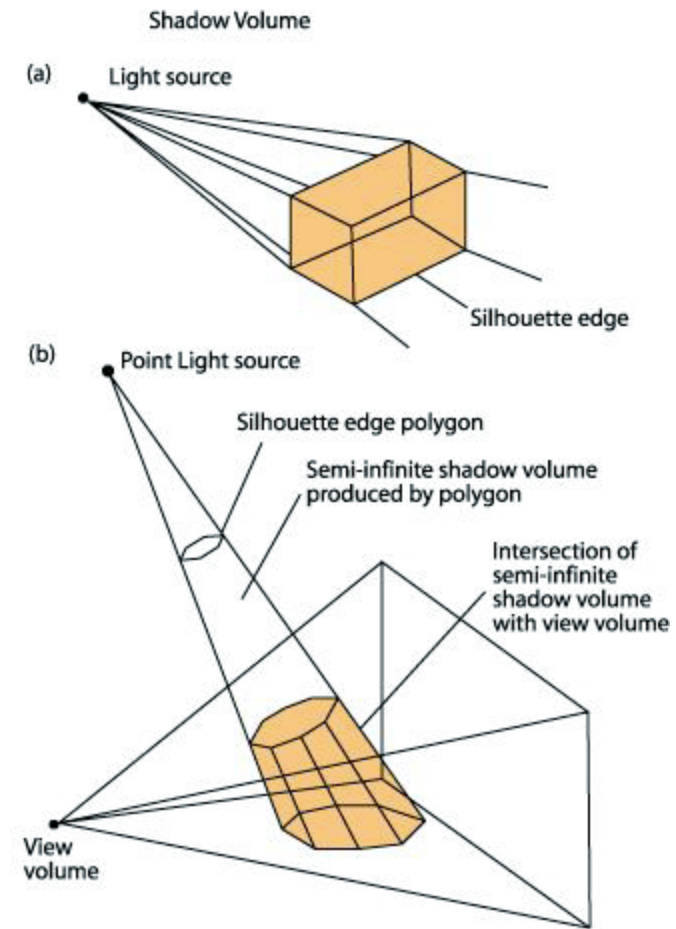
- Shadows are considered as volume in space
- Originally developed by Crow in 1977
- Extension by Brotman and Badler in 1984
- Heidmann introduced the technology for efficient rendering of shadow volumes in 1991



Initial Considerations

- Shadow volume shape depends on shape and position of light source and shadow-caster
- All points inside shadow volume are shadowed
- Shadow volume generation
 - Find silhouettes of shadow-caster regarding light source
 - Extrudes silhouettes in light ray direction to create polygons
 - Connect resulting polygons to obtain shadow volume

Shadow Volume Geometry



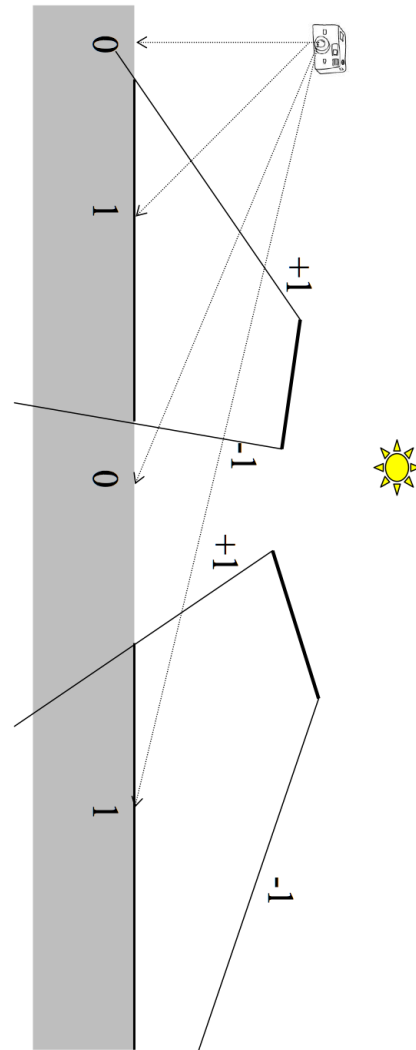
Shadow Volume Realization

- Silhouettes edges can be found by analyzing normals
 - Silhouette edge lies between front and back face (as seen from light source)
- Silhouette edges are stored as vertex list containing vertices V
- For convex shadow-casters
 - Add new (infinitely far-away) vertices for each line between L and V
 - Clip shadow volume against view volume

Shadow Volume Properties 1/2

- All points inside shadow volume are shadowed
- Shadowing can be determined for each view ray by intersection tests
 - Assumption: Camera not in shadow volume
 - At each intersection with front-face shadow polygon, intersection counter s_{cnt} is incremented
 - At each intersection with back-face shadow polygon, intersection counter s_{cnt} is decremented
 - Positions on view rays with $s_{cnt} = 0$ are illuminated, all others are shadowed
- Only intersections before point to be rendered need to be considered

Shadow Volume Properties 2/2



Stencil Shadow Volume

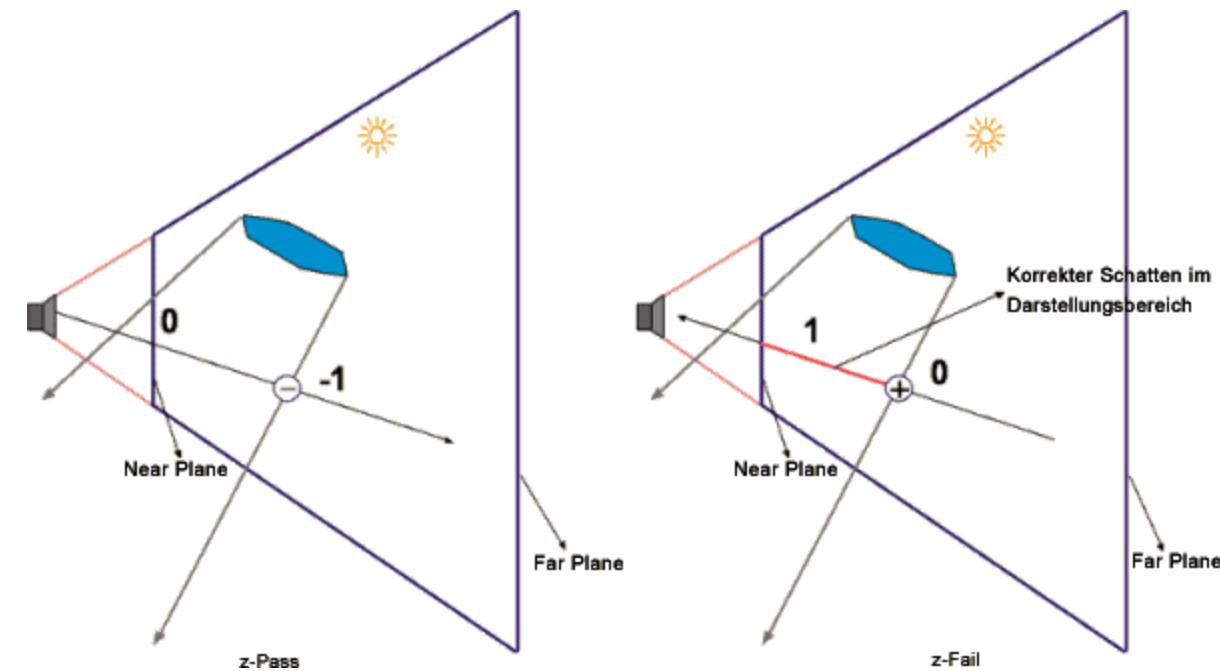
- Generation of shadow volumes per frame
 - Works well for scenes with few moving objects
 - Expensive for complex shadow volumes
 - Improvement by simplifying shadow-casting geometries
- Four rendering passes required
 - Render scene exclusively with ambient lighting
 - Render front-face shadow polygons and increment stencil buffer
 - Render back-face shadow polygons and decrement stencil buffer
 - Render parts of scene that are not in shadow (stencil buffer equals 0)
- However, some details have to be considered!

Implementation

- Determine visible scene objects
 - Render scene without writing in color buffer
- Represent shadow volume in stencil buffer
 - Initialize stencil buffer with 0
 - Render front-face shadow polygons and increment stencil buffer by 1
 - Render back-face shadow polygons and decrement stencil buffer by 1
 - => Shadowed image regions are marked by >0 in stencil buffer
- Render shadowed parts of shadow-receiving scene objects
 - Render only where stencil value >0
- Render non-shadowed parts of shadow-receiving scene objects
 - Render only where stencil value $=0$

Special Case: COP in the shadows

- When COP lies in a shadow volume, the situation can be unclear
 - Near - or far -plane can intersect shadow volume
- Possible solutions
 - Test COP against all shadow volumes $s_{cnt} = 0$
 - Clip shadow volume against view volume and perform capping



Shadow Volume Example



Shadow Volume Conclusions

- Advantages
 - Shadow-casting and shadow-receiving objects can have arbitrary shapes
 - Applicable for multiple light sources
 - Level-of-detail adaptable through approximated geometries
 - Shadow volumes for static pairs of objects and lights can be precalculated
 - Self-shadowing possible
- Disadvantages
 - Many new polygons need to be rendered
 - Long thin triangles are very expensive in rasterization
 - Bit depth of the stencil buffer may not be sufficient for very complex scenes (buffer overflow)
 - Scene objects must be closed (no holes)

10.5 Shadow Maps

Depth test from the point of view of the light source



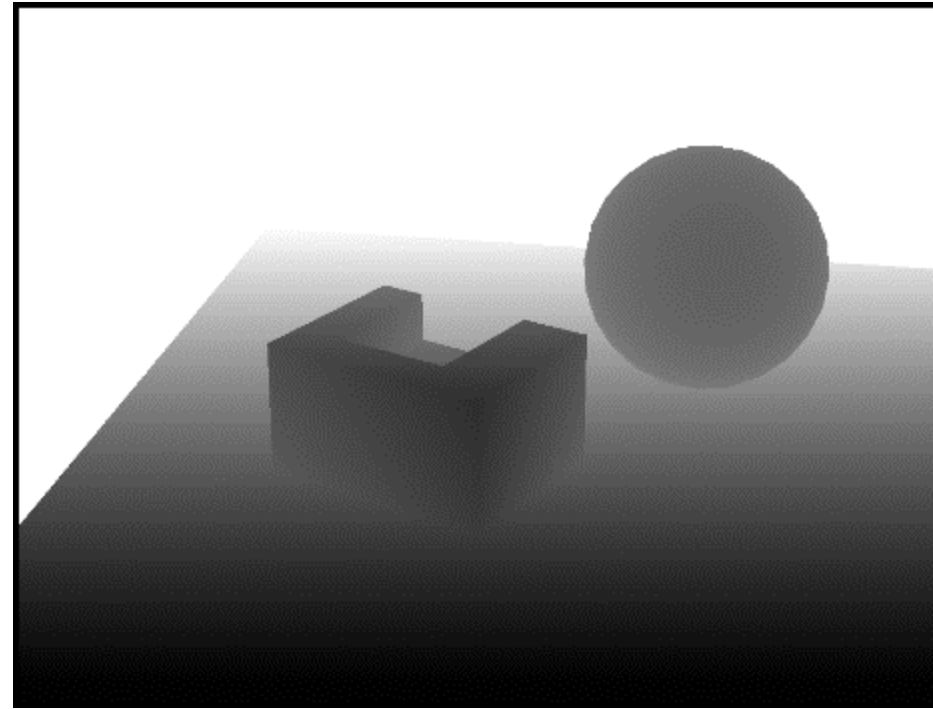
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Shadow Mapping

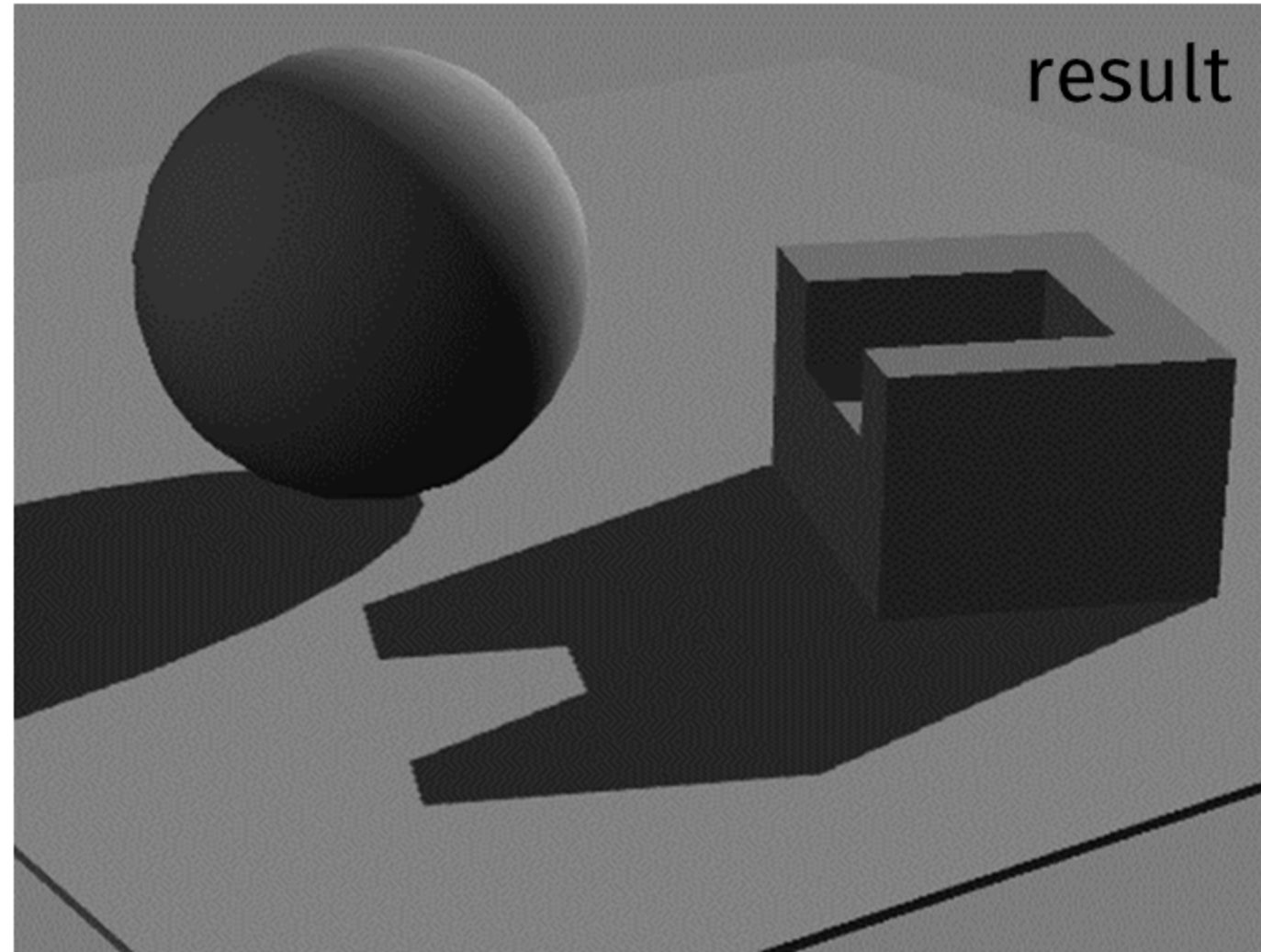
- Based on z-Buffer (contains depth relative to the light source)
- First used within Pixar's Renderman



Projected Shadows

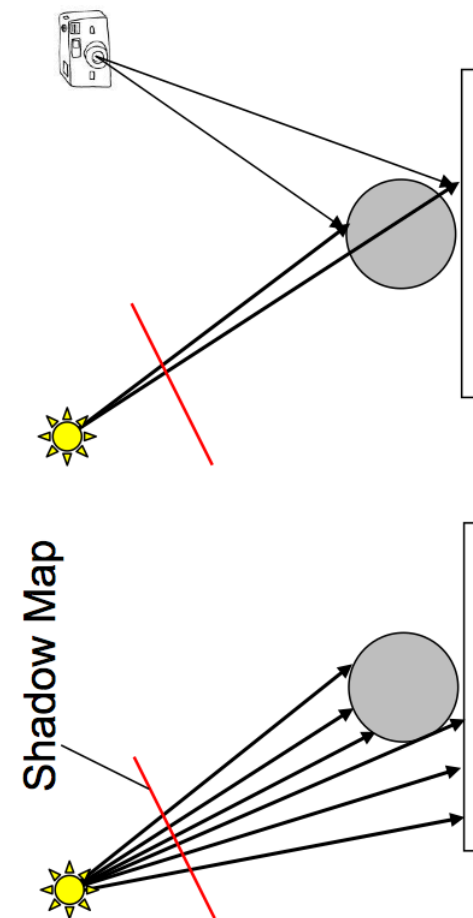


- depth texture
- Rendering algorithm
 - Render shadow-casting objects as seen from the light source into depth texture
 - Project depth texture using projective texturing when rendering shadow-receiving objects
- Characteristics of the process
 - Works for arbitrarily shaped scene objects
 - Shadow-casting and shadow-receiving scene objects can be separated
 - Self-shadowing becomes possible

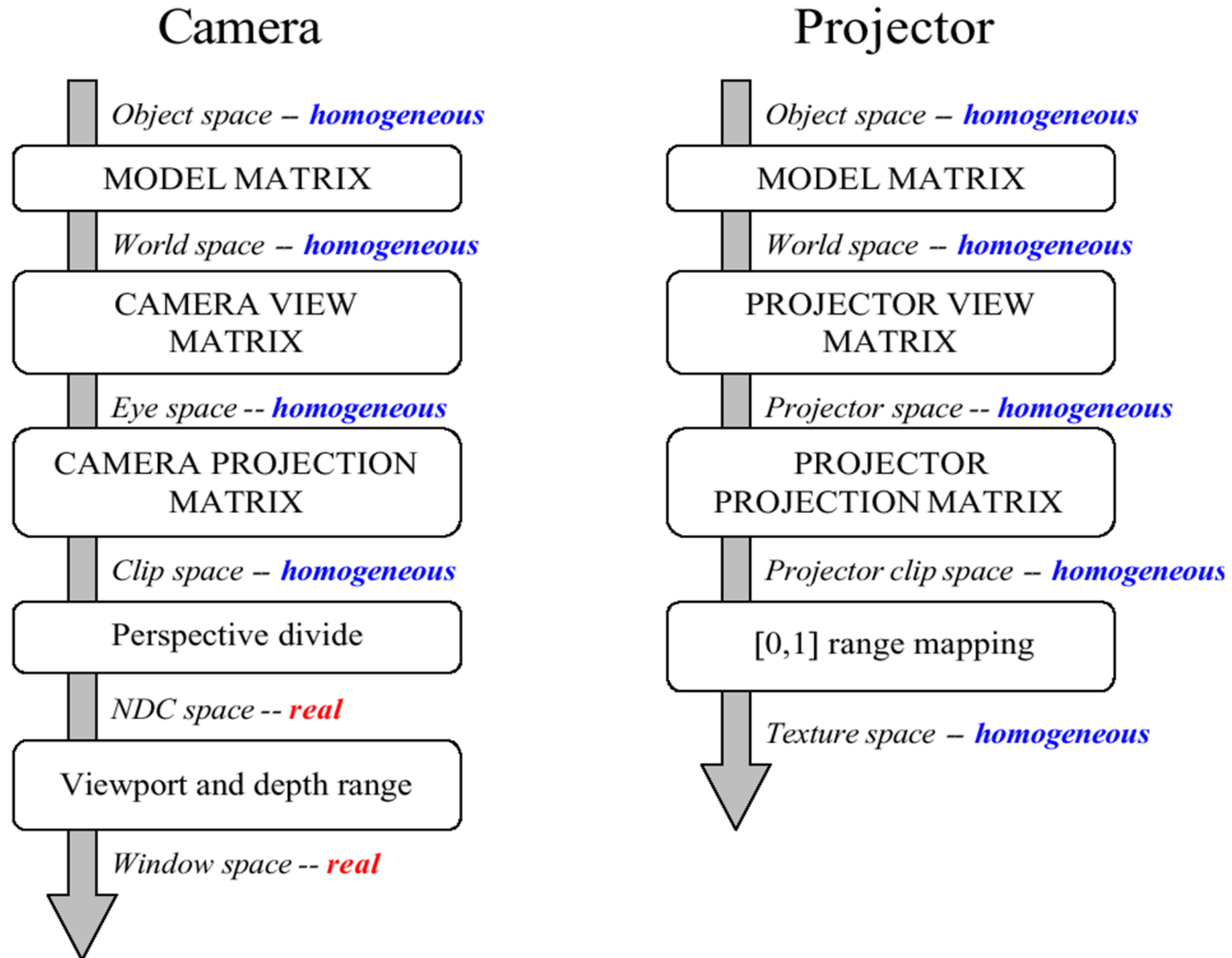


Depth Comparison

- For each vertex V in world coordinates determine associated depth value in shadow map
 - Transform V into camera's eye space
 - Use light source's 4x4 transformation matrix to transform V into light's clip space
 - Compare depth value with associated depth value in shadow map



Coordinate Systems



Pseudo Code

```
// generate shadow map w.r.t light source
generateShadowMap();

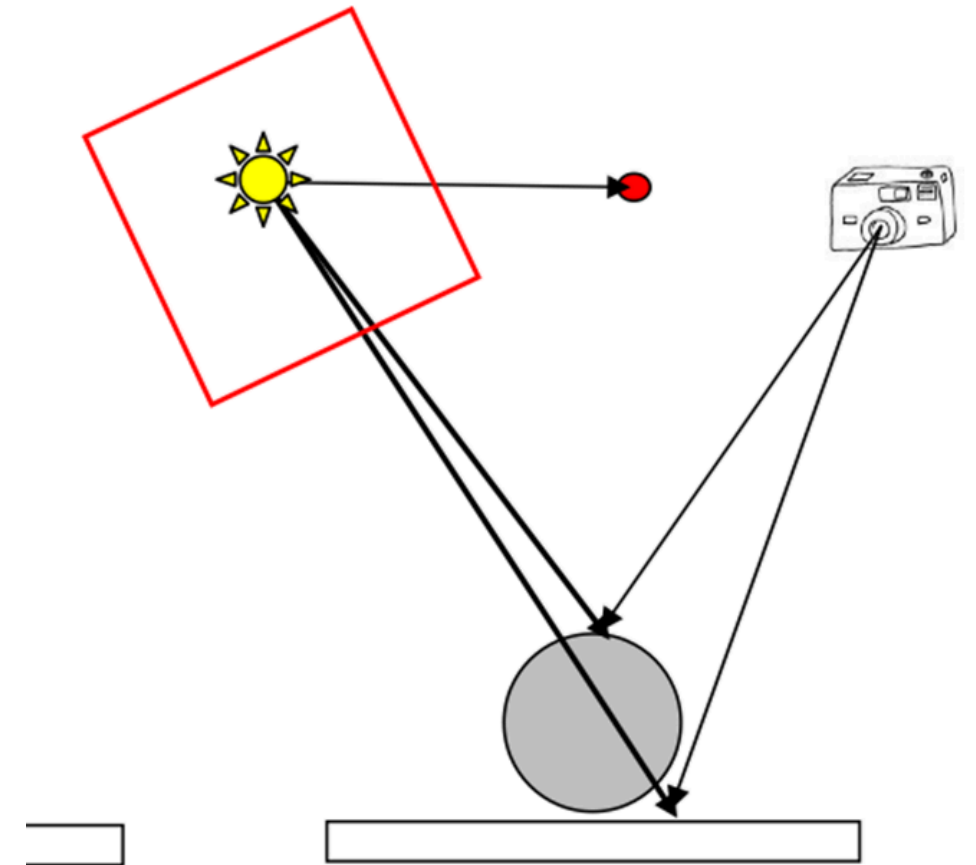
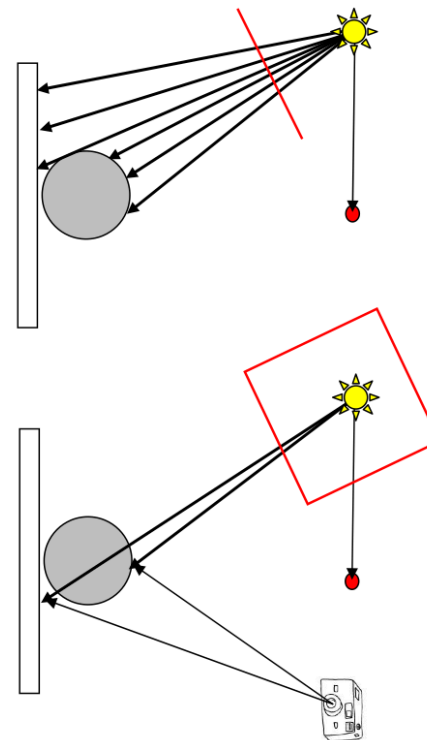
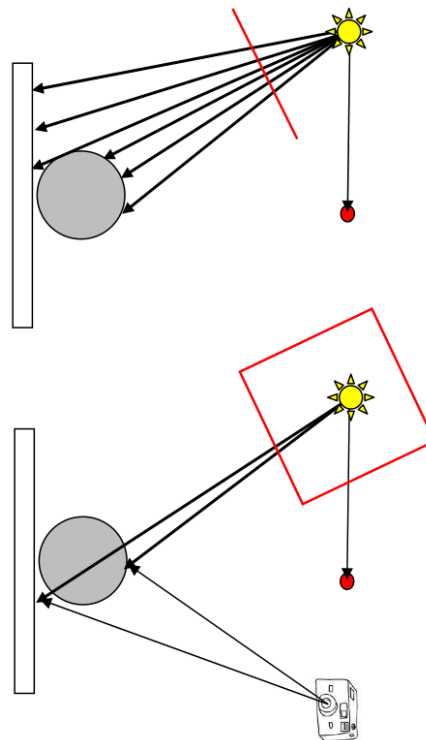
// render scene from COP
for all rasterized fragments do {
    Transform fragment xyz into lights coordinate system
    A = depthmap(x,y)
    B = z-value of fragments xyz
    if (A < B)
        fragment is shadowed
    else
        fragment is lit
    }
}
```

Properties

- Shadow maps are static when scene and lighting are static
- Shadow quality depends on shadow map resolution
- Shadow mapping has three main problems
 - View volume problem
 - Bias problem
 - Aliasing problem

View Volume Problem

- When vertex lies outside light source's view volume, required depth information is missing
- Possible remedies
 - Use cubic shadow map (similar to cubic environment map)
 - Use only spotlights



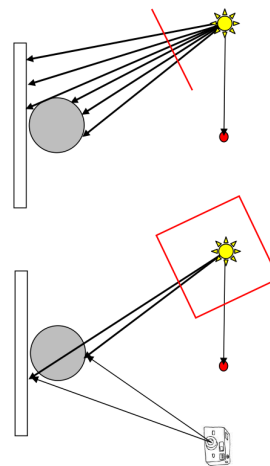
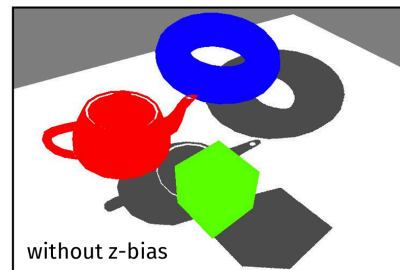
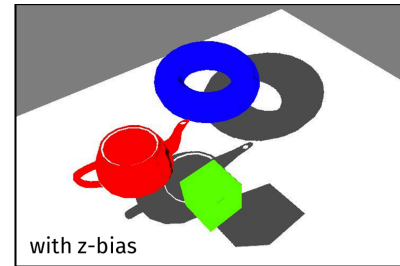
Bias Problem 1/2

Incorrect self-shadowing occurs for
all fragments with $\text{ShadowMap}(x',y') \approx z'$
Can be corrected by
increasing depth precision of z-buffer
applying z-bias

or
 $\text{ShadowMap}(x',y') \approx z'$

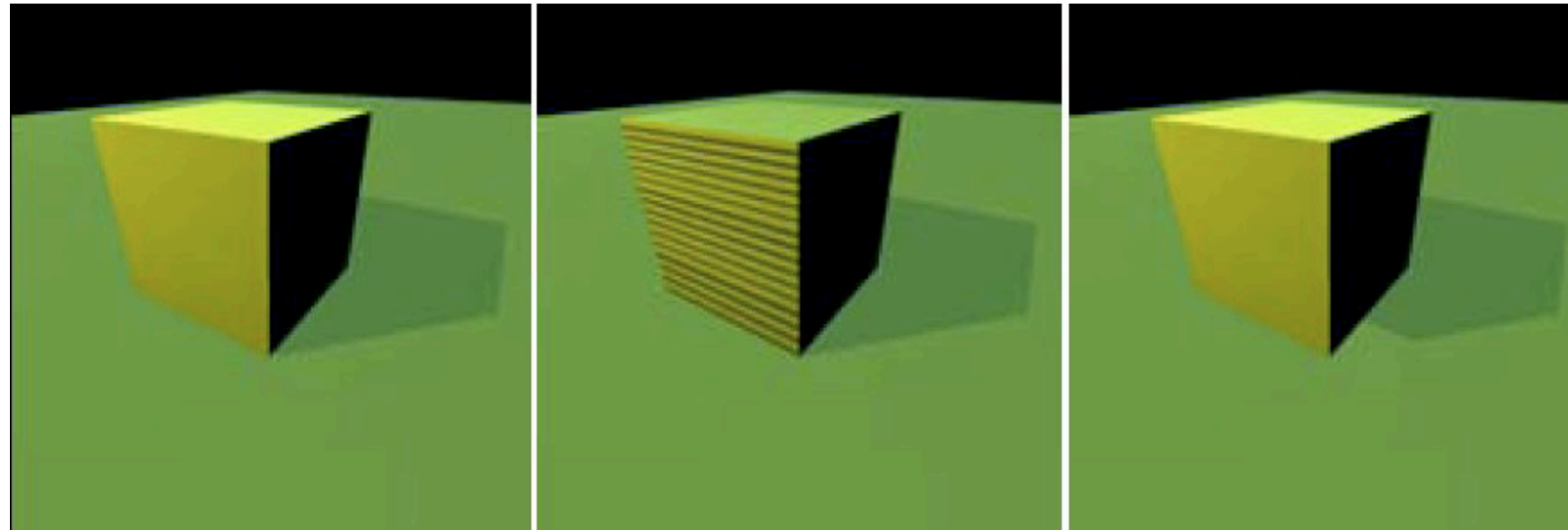
z-buffer

□



Bias Problem 2/2

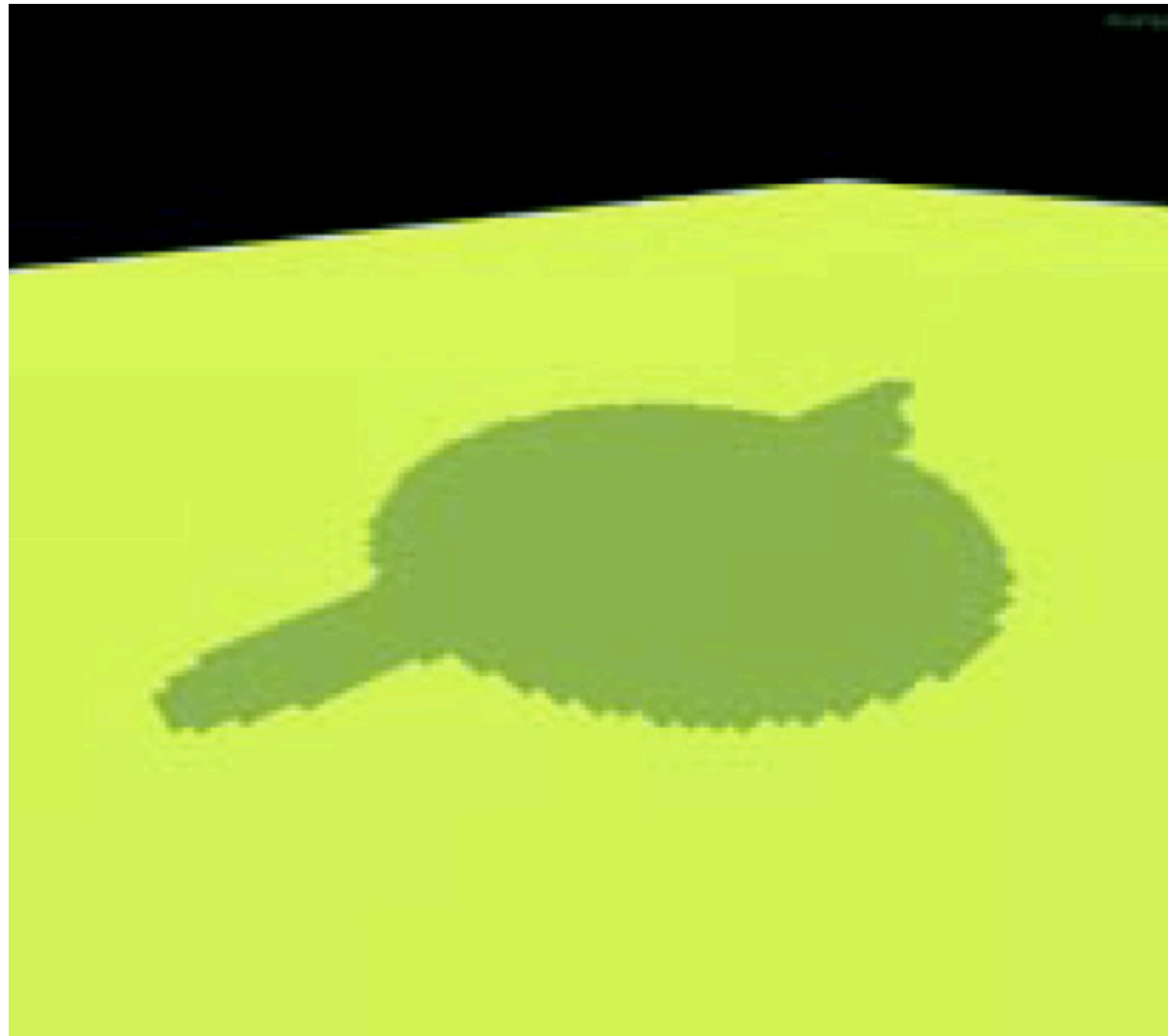
It is difficult to choose a good z-bias
 $\text{ShadowMap}(x',y') + \text{bias} ?$



- good
- too low
- too high

Aliasing Problem 1/3

- Aliasing occurs when shadow map is undersampled(i.e., a texel in shadow map covers several pixels)
- Particularly bad when camera and light source are facing each other



Aliasing Problem 2/3

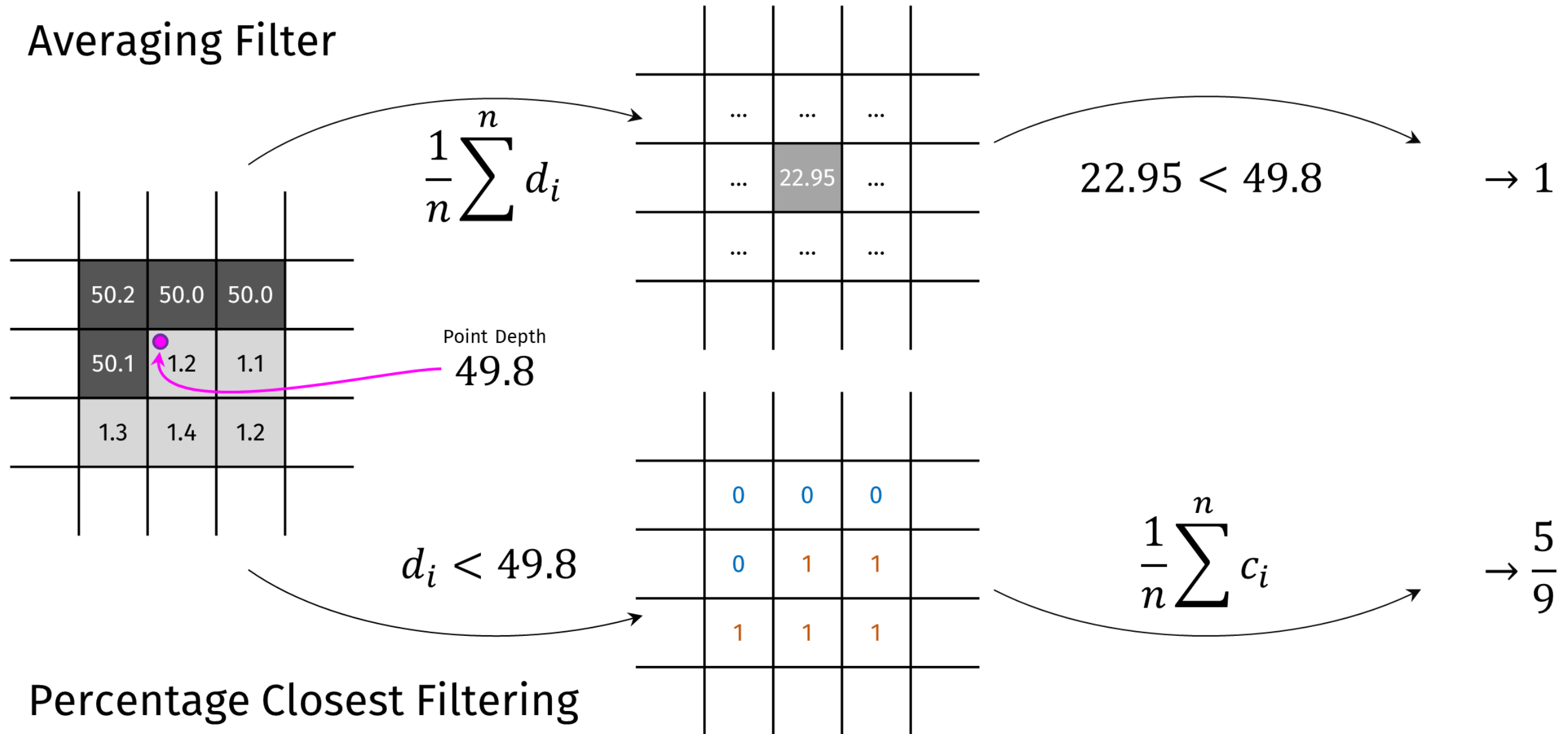
- Filter (= weighted average formation over neighbors) of a depth value does not make sense
- Percentage Closest Filtering (PCF)
 - Filter results of shadow tests (weighted average of comparison results)
 - Bias selection is made even more difficult
- Choosing the filter kernel
 - 5x5 regions generate antialiased shadows
 - Larger filter regions can be used to simulate soft shadows



Aliasing Problem 3/3

- Averaging Filter

Averaging Filter



- Percentage Closest Filtering

Chapter Summary

- Shadows in Visual Perception
 - Enhance depth perception and object relationships
 - Create appealing visual effects and atmosphere
- Shadow Definitions and Types
 - Shadow-casting (Occluder) vs. shadow-receiving (Receiver) objects
 - Self-shadowing: Same object casts and receives shadow
 - Hard shadows (umbra) from point lights, soft shadows (penumbra) from area lights
- Shadow Calculation Methods
 - Techniques include ground plane shadows, light maps, shadow volumes, and shadow maps
 - Ground Plane Shadows
 - Static and dynamic shadow polygons
 - Challenges with representing dynamic scenes and soft shadows
- Light Maps
 - Precomputed static shadows with interpolation for soft shadows
- Shadow Volumes
 - Geometrically enclosed shadows

References

Literature and other sources used in this chapter



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References

- Text Books
 - T. Akenine-Möller, E. Haines: Real-Time Rendering, Addison-Wesley. (Chapter 6: Shadowing)
 - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer Graphics (3rd ed. Edition), AK Peters 2009. (Chapter 10.7: Shadow Maps)
- Research Publications
 - Crow, F., Shadows Algorithms for Computers Graphics, SIGGRAPH 1977.
 - Lance Williams: Casting curved shadows on curved surfaces, SIGGRAPH 1978.